28. Logging: journald, syslog, and log files

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Phase: Phase 10 — Observability and troubleshooting
Estimated time: 1–2 days
Slug: `28-logs-rsyslog`

Learning objectives

- Find logs under /var/log
- Use journalctl effectively
- Rotate logs conceptually
- Correlate time across hosts

Prerequisites

Complete prior numbered lessons. Prefer a real Linux VM/container for Linux-specific topics.

```
sudo tail -f /var/log/syslog   # Debian-like
sudo journalctl -xe
sudo journalctl -u ssh -S today
```

Log rotation (logrotate) prevents disks filling. Centralize logs in real infra (Loki/ELK) — but master local logs first.

Keep clocks synced (chrony/ntp) or timelines lie.

Common mistakes

- Deleting active log files to free space the wrong way
- No retention policy
- Ignoring permission on log dirs

Exercises

Work in `exercises/28-logs-rsyslog/`. Run:

```
./test.sh
```

- **A (Drill):** make `./test.sh` pass
- **B (Apply):** note one real command you ran on a Linux host related to this lesson
- **C (Challenge):** rewrite your solution from memory the next day

Do not open `solutions/28-logs-rsyslog/` until you have an honest failing attempt.

Self-check

1. Where are traditional text logs?
2. journalctl -u meaning?

3. Why NTP for logs?

Further reading

- https://www.freedesktop.org/software/systemd/man/journalctl.html